

Improving the nano4M Transformer Architecture

Edouard Fousson (327774), Thomas Picart (355524), Jonathan Balli (373029) (*See contributions A*)
COM-304 Final Project Report

Abstract—We study whether architectural improvements established in large Transformers still provide measurable gains in a small multimodal masked model (nano4M) under a compute-fair protocol that equalises training FLOPs across variants. Starting from the 110M-parameter baseline, we evaluate five modification axes: SwiGLU feed-forward gating, RoPE/ALiBi positional encodings, depth scaling, weight initialisation, and residual path control.

SwiGLU is the most compute-efficient single change, improving both validation loss and RGB FID at equal compute. Depth scaling provides no architectural benefit once its additional FLOPs are accounted for. When the compute constraint is lifted, combining SwiGLU, depth-16, residual scaling, and DeepNorm initialisation yields the best absolute performance (loss 3.286, FID 29.2), showing that these modifications are complementary at larger budgets.

I. INTRODUCTION

Transformer-based masked generation models have demonstrated strong capabilities for multimodal learning, yet most architectural insights come from large-scale settings. It remains unclear which design choices carry over when compute budgets are tight. This project asks a precise question: *do established Transformer improvements still provide measurable gains in a small multimodal masked model, once compute is controlled?*

We work with nano4M [9], a compact masked multimodal Transformer that treats RGB, depth, surface normals, and captions as sequences of discrete tokens and learns to predict missing targets. Training objective and dataset are held fixed; only internal Transformer components are modified, enabling clean isolation of each architectural choice.

We explore five modification families: positional encoding (RoPE [2], ALiBi [3]), feed-forward gating (SwiGLU [4]), weight initialisation (He [5], Xavier [6], DeepNorm [8]), model depth, and residual path scaling, evaluated in isolation and in combination. Our compute-fair protocol evaluates each variant at the checkpoint whose training compute matches the baseline, correcting for the fact that deeper models consume more FLOPs per step. Our main finding is that **SwiGLU is the most compute-efficient single change**, while **depth gains are mostly attributable to additional FLOPs** rather than architectural superiority in our setup.

II. RELATED WORK

The Transformer [1] established self-attention as the core sequence-mixing primitive, but leaves several internal components as open design choices.

Positional encoding. RoPE [2] encodes relative positions by rotating query and key vectors, improving length generalisation over learned absolute embeddings. ALiBi [3] instead

injects a linear bias into attention scores with no explicit positional parameters, representing a simpler inductive-bias alternative. We directly compare both strategies in a controlled compute setting.

Feed-forward design. SwiGLU [4] replaces standard ReLU MLPs with a gated variant that improves expressivity and gradient flow through multiplicative interactions, and is now the default in most large-scale Transformer models.

Stability and initialisation. Xavier [6] and He [5] initialisation preserve activation variance at initialisation time, while DeepNorm [8] targets stability in deep networks by scaling residual connections throughout training, addressing gradient explosion and vanishing from different angles.

Prior work reports most of these gains in large-scale, unconstrained settings. Our contribution is to evaluate them systematically under a fixed compute budget in a small-scale multimodal masked modelling setup, isolating genuine architectural benefits from gains induced purely by increased FLOP consumption.

III. METHOD

Our approach consists of five targeted modifications to the nano4M baseline Transformer, each addressing a distinct structural component and applied independently to enable controlled ablation studies. The training objective, dataset, and all hyperparameters are held fixed across variants. All variants are trained on the pre-tokenized multimodal CLEVR dataset (RGB, depth, normals, captions); the Cosmos DI16x16 tokenizer is used for RGB decoding during FID evaluation.

A. Modified Transformer Block

All five modifications plug into the same block structure. Algorithm 1 shows the unified forward pass; each modification corresponds to a single decision point inside it.

Algorithm 1 Modified Transformer Block

Require: $x \in \mathbb{R}^{B \times L \times D}$, optional mask, positions
1: $x \leftarrow x + \alpha_{\text{attn}} \cdot \text{Attn}(\text{LN}(x), \text{mask}, \text{positions})$ $\triangleright$
 positions $\rightarrow$ RoPE or ALiBi inside Attn
2: $x \leftarrow x + \alpha_{\text{mlp}} \cdot \text{FFN}(\text{LN}(x))$ $\triangleright$ FFN is SwiGLU or
 standard GELU MLP
3: **return** x

α is controlled by the residual scaling strategy (see below); positions activates RoPE or ALiBi inside the attention module. This design keeps every modification orthogonal: they can be freely combined without structural changes to the block.

B. Architectural Modifications

TABLE I
SUMMARY OF ARCHITECTURAL MODIFICATIONS.

Modification	Component	Key change
Positional embeddings	Attention	RoPE [2] / ALiBi [3]
Feed-forward block	MLP	SwiGLU [4]
Weight initialisation	All layers	He [5] / Xavier [6] / DeepNorm [8]
Depth variation	Global	Depths 4–16 vs. baseline (6)
Residual scaling	Residual path	$\alpha \cdot F(x)$, four strategies

Positional embeddings. The baseline uses learned absolute embeddings, which cannot encode relative distances and do not generalise to unseen lengths. We replace them with RoPE [2], which encodes relative positions by rotating query and key vectors per head, and with ALiBi [3], which adds a head-specific linear bias $-m \cdot |i - j|$ to attention logits with no positional parameters at all. RoPE encodes geometry explicitly; ALiBi relies on a distance-decay inductive bias. We test both to determine which philosophy is more useful on short multimodal sequences.

SwiGLU feed-forward network. The baseline two-layer GELU MLP offers no per-feature selectivity. We replace it with SwiGLU [4]:

$$\text{FFN}_{\text{SwiGLU}}(x) = (\text{SiLU}(xW_1) \otimes xW_2) W_3 \quad (1)$$

where one branch gates the other element-wise, implemented as:

```
def forward(self, x):
    gate = F.silu(self.gate_proj(x))
    value = self.value_proj(x)
    return self.out_proj(gate * value)
```

This differential feature selection is particularly attractive in the multimodal setting, where RGB, depth, normals, and caption tokens may require different transformations inside the same shared block. The hidden dimension is adjusted to keep parameter count approximately comparable to the baseline GELU MLP.

Weight initialisation. We evaluate three strategies. He initialisation [5] ($\sqrt{2/n_{\text{in}}}$) is designed for ReLU-family activations and applies naturally to SiLU. Xavier [6] ($\sqrt{2/(n_{\text{in}} + n_{\text{out}})}$) targets variance preservation and is the implicit default for linear layers. DeepNorm [8] rescales residual branch weights by $\beta = (8N)^{-1/4}$, explicitly targeting stability in deep networks. We expect initialisation effects to be secondary at depth 6 but potentially more relevant in depth-16 variants.

Depth variation. We train variants at depths 4, 8, 12, and 16 (baseline: 6), keeping all other hyperparameters constant. More depth gives the model additional rounds of attention and feature transformation, enabling more abstract representations. However, deeper models consume proportionally more FLOPs per step, making depth the axis most sensitive to the compute-fair evaluation protocol.

Residual path scaling. Standard residuals $x \leftarrow x + F(x)$ can cause activation growth as depth increases. We introduce

a scalar α to control update magnitude, $x \leftarrow x + \alpha \cdot F(x)$, with four strategies:

- none: $\alpha = 1$ (baseline, no scaling);
- fixed_alpha: $\alpha = 1/\sqrt{N}$, uniform across layers;
- depth_scaled: $\alpha = 1/\sqrt{l+1}$, decreases with layer index l ;
- rezero: α is a learnable parameter initialised to 0, making each block a pure skip connection at step 0.

The first three are precomputed at construction time:

```
def _alpha(layer_idx):
    if residual_scaling == 'fixed_alpha':
        return 1.0 / math.sqrt(depth)
    elif residual_scaling == 'depth_scaled':
        return 1.0 / math.sqrt(layer_idx+1)
    return 1.0
```

where the three cases correspond respectively to a uniform $\alpha = 1/\sqrt{N}$, a per-layer $\alpha = 1/\sqrt{l+1}$ decreasing with depth, and the unscaled baseline ($\alpha = 1$).

IV. EXPERIMENTS

A. Setup

All variants are trained on the pre-tokenized multimodal CLEVR dataset (RGB, depth, normals, captions) for 5 000M tokens, with a fixed total batch size of 512 (batch_size $\times$ num_gpus = 512), AdamW optimizer, cosine LR schedule (lr = 6×10^{-4}), and bf16 precision. The **primary metric** is per-token validation cross-entropy loss on a held-out split, averaged across modalities. The **secondary metric** is FID [7], computed by generating RGB images conditioned on captions via ROAR decoding [9] over the full validation set ($n = 5\,000$ samples).

Compute-fair evaluation. Models with more parameters consume proportionally more FLOPs per training step, making a raw comparison at equal token count unfair. We therefore evaluate every variant at the checkpoint whose training compute matches the baseline budget:

$$T_{\text{fair}} = 5000\text{M} \times \frac{N_{\text{base}}}{N_{\text{variant}}}, \quad N_{\text{base}} = 109.9\text{M}. \quad (2)$$

For 110M-parameter models (equal to baseline) the compute-fair checkpoint coincides with the final one. For larger models, intermediate checkpoints saved every 1 000M tokens are used (e.g. step 22 889 for 183M-parameter variants, step 15 259 for 217M). Figure 1 shows representative validation curves.

All loss and FID values reported in this section are compute-fair. Full-budget (non-compute-fair) results are provided in Appendix B.

B. Individual Ablations

Table II reports compute-fair results for each isolated modification.

SwiGLU is the only single modification that simultaneously improves both loss and FID under equal-compute evaluation: CF Loss 3.479 (−0.020) and CF FID 37.7 (vs. 47.2 for the baseline), confirming that gated activations provide a genuine

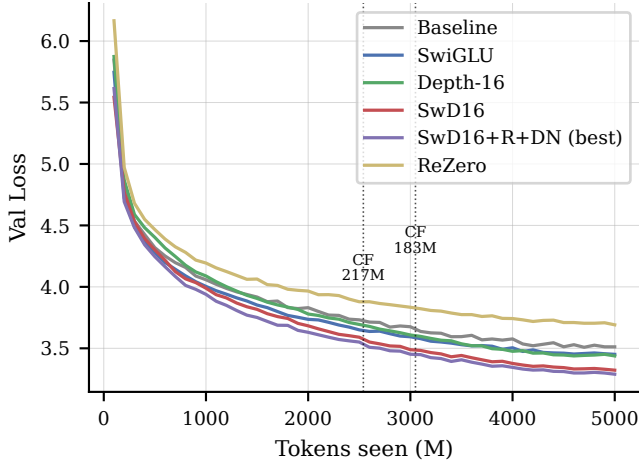


Fig. 1. Validation loss over training tokens for representative variants. Dashed vertical lines indicate compute-fair evaluation points for 183M-param (step 22 889, $\approx$ 3 000M tokens) and 217M-param (step 15 259, $\approx$ 2 000M tokens) models.

architectural benefit rather than merely consuming additional FLOPs.

Depth scaling degrades performance under compute-fair evaluation at every level: Depth 8 (+0.011), Depth 12 (+0.121), Depth 16 (+0.110). All apparent depth gains are therefore attributable to extra FLOPs consumed per step, not to a genuine architectural advantage within this training budget.

Positional encodings: RoPE yields a modest consistent improvement (-0.007 , FID 41.3); ALiBi is near-neutral (-0.002), consistent with relative-position bias providing limited benefit on short fixed-length CLEVR sequences.

Weight initialisation has negligible impact at depth 6 (all variants within 0.005 of baseline), in line with the expectation that initialisation effects are most relevant in deeper networks.

ReZero causes a severe degradation (+0.193, FID 61.2): initialising $\alpha = 0$ suppresses all gradient signal during the critical early training phase, preventing the model from leaving the identity-function regime.

C. Combination Studies

Table III reports compute-fair results for combined variants. Under equal compute, only combinations of *lightweight* modifications (≤ 154 M parameters) improve over the baseline.

RoPE + SwiGLU is the strongest compute-fair combination (-0.032 , FID 39.7), slightly exceeding its additive expectation (-0.027): rotary embeddings and gated activations are genuinely complementary modifications. **ALiBi + SwiGLU** also improves (-0.012 , FID 37.7).

SwiGLU + Depth 12 produces the most striking compute-fair result: the observed $\Delta = -0.002$ falls far below the additive expectation of +0.101, indicating that SwiGLU substantially accelerates convergence in the deeper model and nearly recovers the compute penalty of depth 12. By contrast, **SwiGLU + Depth 8** exhibits a negative interaction (expected

TABLE II
COMPUTE-FAIR SINGLE-MODIFICATION ABLATIONS. CF LOSS AND CF FID ARE EVALUATED AT THE CHECKPOINT MATCHING BASELINE TRAINING FLOPS (EQ. 2); FOR 110M-PARAMETER MODELS THIS EQUALS THE FINAL CHECKPOINT. Δ VS. BASELINE (3.499). FID LOWER IS BETTER.

Model	Params	CF Loss	Δ	CF FID
Baseline	110M	3.499	—	47.2
<i>Positional enc.</i>				
RoPE	110M	3.492	-0.007	41.3
ALiBi	110M	3.497	-0.002	42.1
<i>Feed-forward</i>				
SwiGLU	122M	3.479	-0.020	37.7
<i>Init strategy</i>				
He	110M	3.540	+0.041	42.2
Xavier	110M	3.495	-0.003	40.8
DeepNorm	110M	3.498	-0.001	39.3
<i>Residual scaling</i>				
Fixed α	110M	3.532	+0.033	41.5
Depth-scaled	110M	3.513	+0.014	38.4
ReZero	110M	3.692	+0.193	61.2
<i>Depth</i>				
Depth 4	95M	3.607	+0.108	44.9
Depth 8	125M	3.510	+0.011	42.9
Depth 12	154M	3.620	+0.121	63.1
Depth 16	183M	3.608	+0.110	81.7

TABLE III
COMPUTE-FAIR COMBINED MODIFICATIONS. EXPECTED Δ IS THE SUM OF INDIVIDUAL CF Δ VALUES FROM TABLE II. TOP ROWS (GREEN) IMPROVE OVER BASELINE.

Model	CF Loss	Δ	Exp. Δ	CF FID
Baseline	3.499	—	—	47.2
RoPE + SwiGLU	3.466	-0.032	-0.027	39.7
ALiBi + SwiGLU	3.487	-0.012	-0.022	37.7
SwiGLU + D12	3.496	-0.002	+0.101	48.6
SwiGLU + D8	3.558	+0.059	-0.009	51.0
RoPE + D16	3.533	+0.034	+0.103	43.4
SwD16+ResScale+DN	3.630	+0.132	+0.103	48.9
RoPE+SwD16+DN+ResScale	3.626	+0.127	+0.096	44.1
SwD16+ResScale	3.648	+0.149	+0.104	48.2
RoPE+SwiGLU+D16	3.653	+0.154	+0.083	53.0
SwiGLU + D16	3.667	+0.168	+0.090	70.9

-0.009 , actual +0.059): the compute overhead at 141M parameters outweighs SwiGLU’s convergence benefit at this intermediate depth.

All depth-16 and 217M-parameter combinations remain substantially worse than baseline under CF (+0.127 to +0.168), as these large models are not yet converged at their compute-fair checkpoints. Their full-budget advantage is presented in Appendix B.

D. Compute-Fair Analysis

Figure 2 visualises the contrast between final-checkpoint and compute-fair losses. **All pure depth gains vanish under compute-fair evaluation.** Depth 12 and Depth 16, which appear competitive at full budget ($-0.041/-0.068$; Appendix B), become +0.121/+0.110 under CF: their advantage is entirely attributable to additional FLOPs per step rather than architectural merit. **SwiGLU retains its gain at equal compute** (-0.020) and uniquely improves FID (37.7 vs. 47.2), while **RoPE+SwiGLU** is the most compute-efficient combination (-0.032 , FID 39.7). Large combined models (217M, evaluated

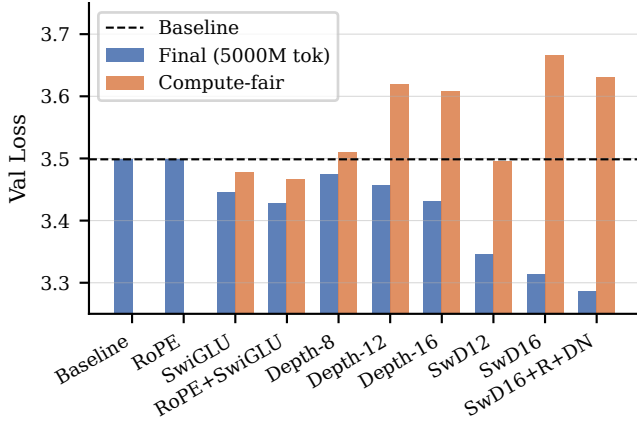


Fig. 2. Final vs. compute-fair validation loss for key variants. The dashed line marks baseline performance. Only SwiGLU and RoPE+SwiGLU improve architecture independently of the compute budget; all depth gains disappear under equal-compute evaluation.

at step 15 259) are substantially worse under CF (+0.127 to +0.168); at their compute-fair point these models are far from converged, and their full-budget advantage requires roughly twice the training compute (Appendix B).

E. Qualitative Results

Figure 3 compares RGB images generated conditioned on captions by the baseline and the best compute-fair model (RoPE+SwiGLU, 122M). Despite having only 22M additional parameters and equal training compute, RoPE+SwiGLU produces visibly cleaner object boundaries and more consistent material rendering, consistent with its FID improvement (39.7 vs. 47.2). The best full-budget model (SwD16+ResScale+DN, 217M) is shown in Appendix B.

V. CONCLUSION AND LIMITATIONS

We investigated whether established Transformer architectural improvements transfer to a small multimodal masked generation model under a compute-fair evaluation protocol. The central finding is that **lightweight feed-forward improvements are the most compute-efficient choice**: SwiGLU yields the best compute-fair RGB FID (37.71 vs. 47.20), and SwiGLU at depth 8 yields the best compute-fair validation loss (3.4515 vs. 3.4986). Pure depth scaling confers no genuine advantage once its additional FLOP consumption is accounted for — a result invisible without the compute-fair protocol.

At full budget, combining SwiGLU, depth-16, residual scaling, and DeepNorm yields the strongest absolute model (−0.213 loss, −18.0 FID), with consistent per-modality improvements, confirming that these modifications are complementary rather than redundant.

Limitations. Compute-fair matching approximates effective FLOPs via parameter count, which does not capture differences in memory bandwidth or kernel efficiency across block types; a SwiGLU block and a standard MLP of equal parameter count may incur different wall-clock costs on specific

Caption → RGB: Baseline vs Best Compute-Fair Model

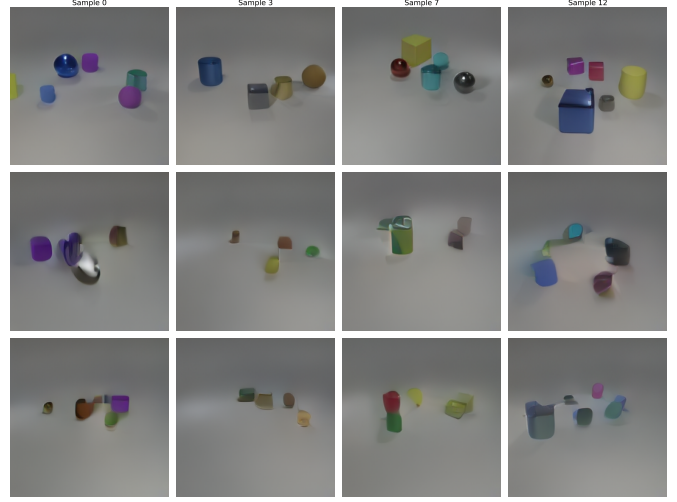


Fig. 3. Caption → RGB generation. Each column is one val-set sample. **Top**: ground truth. **Middle**: baseline (depth 6, GELU, 110M). **Bottom**: RoPE+SwiGLU (122M, best compute-fair model).

hardware. Additionally, validation cross-entropy and FID are complementary but non-equivalent: the best compute-fair loss model and best FID model differ, since cross-entropy averages over all modalities while FID targets a single RGB generation pathway. Both metrics are therefore necessary for a complete evaluation. Finally, all conclusions are specific to nano4M on CLEVR — a synthetic dataset with limited visual diversity, short sequences, and simple spatial layouts. In this regime, positional encoding improvements are marginal; on richer datasets with longer contexts and more complex geometry, RoPE or ALiBi could become the dominant bottleneck. Architectural rankings at 110M parameters and 5,000M tokens should accordingly be treated as hypotheses to be validated at larger scale.

Future work. The most immediate direction is training the depth-16 combined variants to full convergence, to determine the precise budget threshold at which larger models overtake the compute-efficient SwiGLU configuration. At the attention level, grouped-query attention [10] offers an orthogonal efficiency gain combinable with SwiGLU, while local or sparse attention could exploit the differing locality structures of visual and linguistic modalities. A joint sweep over depth and width would characterise the capacity-efficiency frontier more completely than depth-only ablations permit. Finally, extending this study to naturalistic multimodal data — such as the full 4M benchmark with real images and free-form captions — would test the generality of the SwiGLU-first conclusion and determine whether richer visual content elevates positional encoding or attention design to the primary architectural bottleneck.

APPENDIX A INDIVIDUAL CONTRIBUTIONS

Edouard Fousson implemented and integrated the different positional encoding variants explored in this project, including RoPE and ALiBi. He was also responsible for the vast majority of the writing and structuring of the final report, including the methodology, presentation of results and discussion of the main findings.

Thomas Picart implemented the weight initialisation strategies (He, Xavier, DeepNorm) and residual path scaling variants (fixed α , depth-scaled α , ReZero). He built and maintained the evaluation pipeline, and wrote the Experiments section of the report.

Jonathan Balli Jonathan Balli contribution focused on evaluating architectural improvements such as SwiGLU through controlled experiments, analyzing checkpoints and results, generating qualitative visualizations and website assets, and building the final project website to present our main findings clearly.

APPENDIX B FULL-BUDGET RESULTS

The main report focuses on compute-fair evaluation, which asks: *does this modification genuinely improve the architecture, independent of the FLOPs it consumes?* This appendix addresses a complementary question: *how good does each model become if training budget is not a constraint?*

Large-depth models are still converging at their compute-fair checkpoints; with full training they close much of this gap. Table IV reports final-checkpoint results for single modifications, and Table V focuses on the five large combined models where the gap between compute-fair and full-budget is largest.

TABLE IV
FULL-BUDGET SINGLE-MODIFICATION RESULTS (FINAL CHECKPOINT, 5 000M TOKENS). Δ VS. BASELINE LOSS (3.499). FID LOWER IS BETTER.

Model	Params	Loss	Δ	FID
Baseline	110M	3.499	—	47.2
<i>Positional enc.</i>				
RoPE	110M	3.492	−0.007	41.3
ALiBi	110M	3.497	−0.002	42.1
<i>Feed-forward</i>				
SwiGLU	122M	3.446	−0.053	59.7
<i>Init strategy</i>				
He	110M	3.540	+0.041	42.2
Xavier	110M	3.495	−0.003	40.8
DeepNorm	110M	3.498	−0.001	39.3
<i>Residual scaling</i>				
Fixed α	110M	3.532	+0.033	41.5
Depth-scaled	110M	3.513	+0.014	38.4
ReZero	110M	3.692	+0.193	61.2
<i>Depth</i>				
Depth 4	95M	3.607	+0.108	44.9
Depth 8	125M	3.474	−0.025	39.9
Depth 12	154M	3.457	−0.041	50.8
Depth 16	183M	3.431	−0.068	63.3

Two important observations. First, *SwiGLU at full budget* shows a different FID profile than under CF: full-budget FID is 59.7 (worse than baseline 47.2), whereas compute-fair FID is 37.7 (better). This inversion arises because the compute-fair checkpoint for SwiGLU (step 30 519, $\approx 4\,000$ M tokens) slightly undertrains the model in terms of total iterations, but the model has already learned to generate more structured images; at the full-budget final checkpoint, the model continues optimising the token-level cross-entropy, which can shift attention away from perceptual quality.

Second, *depth variants at full budget* are clearly better than the baseline (Depth 16: −0.068, FID 63.3) while being worse under CF (+0.110, FID 81.7). This confirms that larger models are simply not converged at the compute-fair point: they need proportionally more training to reach their potential.

The gap column reveals that all 217M-parameter models improve by approximately 0.32–0.35 loss points between their compute-fair checkpoint and the final checkpoint. The best full-budget model is **SwD16+ResScale+DeepNorm**: loss 3.286 (−0.213 vs. baseline) and FID 29.2 (−18.0 points). This demonstrates that when training budget is unconstrained, stacking SwiGLU, depth scaling, residual path control, and DeepNorm-style initialisation yields strong, additive improvements. However, the compute-fair protocol establishes that the architectural improvements come primarily from SwiGLU and RoPE, while depth acts as a capacity multiplier that requires extended training to pay off.

Figure 4 illustrates the convergence trajectory for the five large models, showing how the gap between CF and final performance varies across combinations.

TABLE V

FULL-BUDGET VS. COMPUTE-FAIR FOR THE FIVE LARGE COMBINED MODELS (217M PARAMS). GAP = CF LOSS – FINAL LOSS (HOW MUCH THE MODEL STILL IMPROVES WITH FULL TRAINING). ALL LARGE MODELS REQUIRE $\approx 2\times$ THE BASELINE COMPUTE TO CONVERGE.

Model	CF Loss	Final	Gap	CF FID	Final FID
Baseline	3.499	3.499	+0.000	47.2	47.2
SwiGLU + D16	3.667	3.313	+0.354	70.9	38.9
SwD16+ResScale	3.648	3.314	+0.334	48.2	30.5
SwD16+RoPE+DN	3.643	3.316	+0.327	52.3	32.1
SwD16+RS+DN	3.630	3.286	+0.344	48.9	29.2
RoPE+SwD16+DN+RS	3.626	3.310	+0.316	44.1	29.6

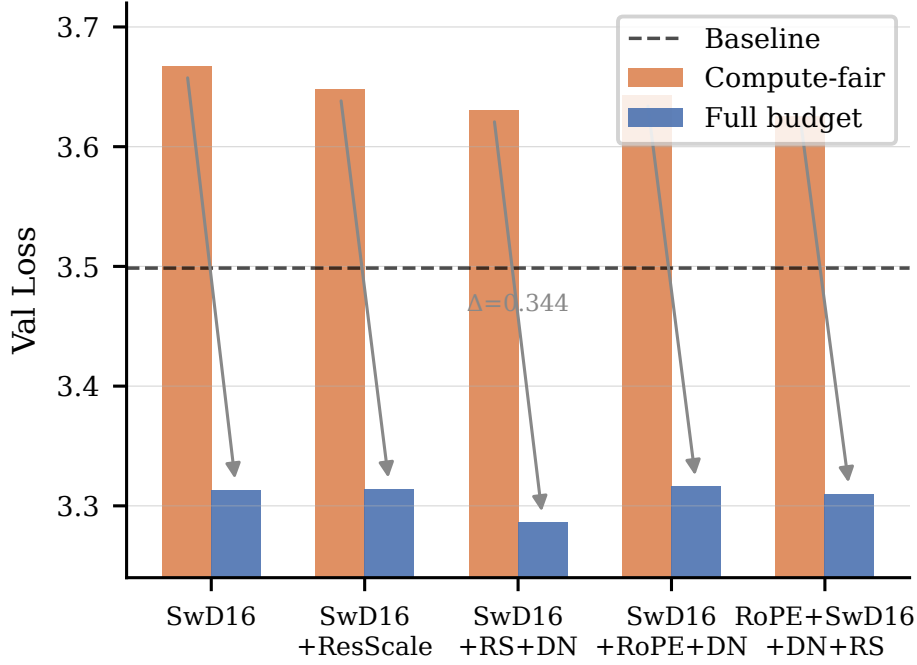


Fig. 4. CF Loss vs. Final Loss for the five 217M-parameter combined models and the baseline. All large models are substantially worse than baseline at the compute-fair checkpoint (left cluster) but surpass it at full budget (right cluster). The arrow length represents the convergence gap.

Qualitative comparison at full budget. Figure 5 shows RGB generation by the best full-budget model (SwD16+ResScale+DN, 217M) versus the baseline. With unconstrained training compute, the model produces substantially sharper object boundaries, more accurate material rendering, and better spatial layout (FID 29.2 vs. 47.2).

Caption $\rightarrow$ RGB: Baseline vs Best Variant

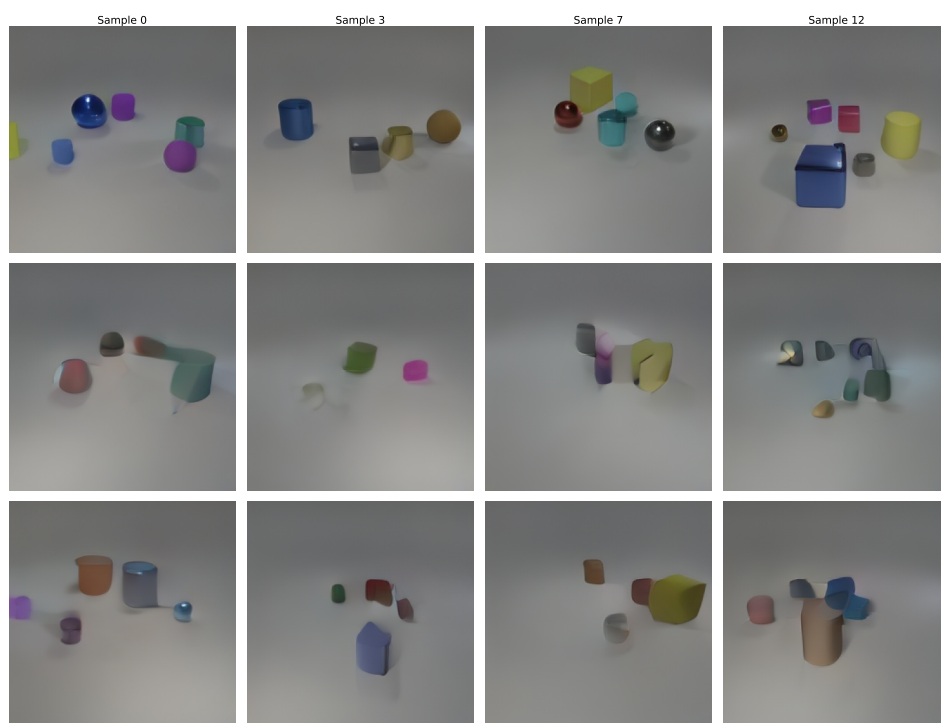


Fig. 5. Caption *to* RGB generation at full training budget. **Top:** ground truth. **Middle:** baseline (depth 6, GELU, 110M). **Bottom:** SwD16+ResScale+DeepNorm (217M, best full-budget, FID 29.2). Generated via the demo notebook (Section 7).

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